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# Internship Assignment: Self-Learning PMS/AIF Statement Parser

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**ACTIVV MONEY** · Software Development Internship · Take-Home Assignment

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## Overview

You will build a service that reads Portfolio Management Service (PMS) and Alternative Investment Fund (AIF) statements from email and turns them into clean, structured data. You are encouraged to use AI coding tools (Claude, ChatGPT, Copilot, Cursor or similar) to build it.

After submission, you will walk us through your solution in a one-on-one discussion. That discussion carries most of the weight. We care less about how much code you shipped and more about how deeply you understood the problem, the choices you made, and the ways your system can fail.

***Why does this matter to us?*** Investors with PMS and AIF holdings receive monthly or quarterly statements from many different providers and custodians. Each one uses its own layout, mostly as PDFs, sometimes as Excel. Turning these into a single consolidated view is tedious, error-prone work today.

## Problem statement

Build an application that watches a mailbox, picks up PMS/AIF statements, and extracts the portfolio data in them into a standard structure.

**The core constraint: AI only for formats it has never seen.** A large language model may be used only the first time the system encounters a statement format it does not recognise. In that first encounter, the LLM's job is to help the system *learn* the format, for example by generating a parser, a template or an extraction configuration. Every later statement in a format the system already knows must be parsed deterministically, with no LLM call at all.

This constraint is the heart of the assignment. Calling an LLM on every statement is slow, costly, and can give different answers for the same input. A learned, repeatable parser avoids that, but it raises hard questions you will need to answer:

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- How does the system recognise that a new statement matches a format it has already learned?
- What exactly does it store when it "learns" a format, and why that representation?
- Without an AI checking the output, how does the system know a deterministic parse is correct?
- What happens when a known provider quietly changes its layout?

## Functional requirements

Your solution must cover the following end to end. You may choose any language, framework and storage.

1. **Email ingestion.** Connect to a mailbox (a Gmail test account is fine) and pick up emails that carry statements. Decide for yourself how the system tells a statement email from any other email.
2. **Document handling.** Read PDF statements at minimum. Excel or CSV support is a plus. Real statements are often password protected, so your design must account for that.
3. **Format recognition.** For each document, decide whether it matches a format the system already knows.
4. **Learning a new format.** For an unknown format, use an LLM once to produce something reusable, then store it so the next statement of that format needs no LLM.
5. **Deterministic parsing.** For a known format, extract data using only the stored, learned artefact.
6. **Standard output.** Produce data in one common structure regardless of provider. At minimum, capture the fields below.
7. **Validation.** Check that each extracted statement is internally consistent, and decide what the system does when it is not.
8. **Review and visibility.** Provide a simple way (CLI, web page or API) to see processed statements, their status, and which ones need human attention.

| Area | Minimum fields to capture |
|---|---|
| Statement | Provider name, strategy or scheme name, client identifier, statement period, currency |
| Holdings | Security name, ISIN where present, quantity, cost, market value, weight |
| Transactions | Date, type (buy, sell, dividend, fee, inflow, outflow), security, quantity, amount |
| Summary | Opening value, closing value, cash balance, fees charged, NAV per unit for AIFs |

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**Stretch goals** (optional, but they make for a richer discussion): detecting when a known provider has changed its layout; handling a single email that carries statements for several clients or strategies; a consolidated view across providers for one client.

## Using AI tools

You may use any AI coding assistant as much as you like. Treat it as a capable but junior colleague: it writes code quickly, but it does not know our domain, and it will confidently produce things that look right and are not.

- **Keep your AI conversation history** and submit it (exported chats, prompt logs or screenshots). We will not mark you down for heavy AI use. We want to see how you steered it.
- **You must be able to explain every part of your code.** In the discussion, "the AI wrote that part" is not an answer. If you cannot explain a piece, say so in your limitations note instead.
- **Do not use real client data.** Create your own sample statements, or anonymise any you obtain. Never commit passwords, PANs, API keys or personal details to your repository.
- **The runtime LLM** (the one your application calls to learn new formats) can be any provider. Document which one, and how your system behaves if that call fails.

## Deliverables

Submit all of the following. The two notes matter as much as the code.

| # | Deliverable | What we expect |
|---|---|---|
| 1 | Source code | A Git repository (GitHub or GitLab link) with a README covering setup, configuration and how to run it |
| 2 | Sample data | At least 3 different statement formats you created or anonymised, plus the output your system produced for each |
| 3 | Known limitations note | Every situation you know your system handles badly or not at all, as specifically as you can |
| 4 | AI usage log | Your AI conversation history, with a short note on where |

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| # | Deliverable | What we expect |
|---|---|---|
| | | the AI helped most and where it misled you |

A specific, honest limitation note is a strength, not a weakness. "May not handle all formats" tells us nothing. A list of concrete failure cases tells us you understood the problem.

## How you will be evaluated

Evaluation happens in a 45 to 60 minute one-on-one discussion after submission. Bring your laptop with the solution running.

During the discussion we will:

- Ask you to walk through your architecture and the reasoning behind your key choices.
- Run your system on **statements you have not seen before**. These are realistic, and some are deliberately messy. Anticipating what real-world statements look like is part of the task.
- Ask you to talk through a change to the requirements and how your design would absorb it.
- Pick parts of your code and ask you to explain them.

| Dimension | What we look for |
|---|---|
| Problem understanding | You grasp why the core constraint exists and what makes it hard |
| Design reasoning | You can explain your choices and their trade-offs, not just describe what you built |
| Correctness and robustness | Your system knows when its own output is wrong, and handles unfamiliar input without silent errors |
| Edge-case thinking | You found failure modes on your own and handled or documented them |
| Code quality | Readable, organised, sensibly tested code that you can explain line by line |
| Use of AI tools | You directed the AI deliberately and caught its mistakes |

A solution that handles fewer formats but is well reasoned will score higher than one that handles many but cannot explain itself.

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## Timeline and submission

| Milestone | Date |
|---|---|
| Assignment issued | 8th October, 2026 |
| Last day for clarification questions | 10th October, 2026 |
| Submission deadline | 17th October, 11:59 PM IST |

Send your submission to **hiring@activv.money** with the subject line *Internship Assignment – [Your Name] – IIT Indore*. Include the repository link, the demo video link and the three notes.

**Ground rules**

- This is an individual assignment. You may use AI tools and public documentation freely, but not help from other people.
- Clarification questions are welcome until the date above. Where the brief is silent, make a reasonable assumption and record it in your design note; how you handle ambiguity is part of what we assess.
- Your repository must stay accessible until the discussion round is complete.

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